

MPV MEDIA PLAYER — COMPREHENSIVE USER GUIDE

1. Overview & Core Capabilities

biraj-mpv-conf is a clean, high-performance configuration suite for mpv. It unites a fluent modern interface, tear-free 4K video playback, fast seekbar hover thumbnails, zero-lag track cycling, and native Windows file dialogs into a single seamless daily driver.

- **Tear-Free Direct3D 11:** Hardware VSync flip presentation (`d3d11-flip=yes`) with borderless fullscreen (`d3d11-exclusive-fs=no`) and display resampling (`video-sync=display-resample`) for locked frame cadence.
- **HDR10 & Dolby Vision:** Real-time tone mapping tuned for 0-frame-drop playback on laptop screens with automatic contrast curve adaptation (`target-contrast=auto`).
- **Debanding & Dithering:** Eliminates color banding artifacts and gradient compression in dark scenes and anime (`deband=yes` , `dither-depth=auto` , `temporal-dither=yes`).
- **Zero-Lag Track Cycling:** 0ms OSD feedback with 25ms anti-spam debouncing for audio and subtitles.
- **Bounded RAM Memory:** Strict 400 MiB forward and 200 MiB backward buffer capping RAM under 1.0 GB.

2. Installation & Directory Structure

Installation Mode	Target Directory Path
Standard (Roaming)	C:\Users\<Username>\AppData\Roaming\mpv\
Portable Mode	C:\<PathToMpv>\portable_config\

3. Video Playback & Audio Pipeline

- **Next-Gen Renderer:** Uses `vo=gpu-next` with libplacebo for clean scaling, gamut mapping, and color fidelity.
- **Safe Hardware Acceleration:** Uses `hwdec=auto-safe` with 16 VRAM buffer frames (< 0.5% CPU load).
- **Audio Buffer Headroom:** 150ms WASAPI lock (`audio-buffer=0.15`) with stream continuity (`audio-stream-silence=yes`) preventing soundbar/DAC sleep and dropouts.
- **Dual Track Shortcuts:** Supports both VLC (`b` / `v`) and standard MPV (`_` / `##` , `j`) cycling with delay sync shortcuts (`Ctrl+[` / `Ctrl+]` , `z` / `Z`).

4. Memory Architecture & Demuxer Cache Safety

Parameter	Configured Value & Behavior
Forward Buffer	<code>demuxer-max-bytes=400MiB</code> (hard read throttle)
Backward Rewind Buffer	<code>demuxer-max-back-bytes=200MiB</code> (FIFO eviction)
Readahead Window	<code>demuxer-readahead-secs=25</code> (absorbs network jitter)
Peak RAM Footprint	~850–950 MB (hard-capped; never climbs indefinitely)
Storage Health	Pure in-memory RAM caching with 0 SSD write wear (<code>cache-on-disk=no</code>)

5. Modern UI, Thumbnails & Automation

- **Dynamic Format Badge** (`hdr_badge.lua`): Auto-announces `DV`, `HDR10+`, `HDR`, `HLG`, `SDR`; toggled with `1`.
- **Seekbar Hover Thumbnails**: `thumbfast` real-time preview thumbnails cached to `\%TEMP%\`.
- **Graceful Auto-Exit** (`auto_exit_eof.lua`): Snappy 4.0s grace period with 2.0s OSD warning (`Exiting...`). Resets watch-later on completion (last 10s threshold) and auto-rewinds to 0:00 on reopen.
- **Clean Resume Notice** (`resume_indicator.lua`): Displays clean OSD notice on resume; ignores starts < 5s.
- **Persistent Pause Badge** (`pause_notify.lua`): Theme-matched pause status with dynamic collision avoidance.
- **Ascending Playlist Sorter**: `sort_playlist.lua` sorts multi-file drops in natural order (`k` / `K`).
- **Unicode File Pickers**: Native Windows dialogs for media (`Ctrl+o`), subs (`Ctrl+s`), audio (`Ctrl+a`).

6. Smart Automation Profiles

Profile Name	Trigger Condition & Behavior
<code>[Window-PiP]</code>	Windowed on-top mode: strips window borders and scales OSC by 1.8x
<code>[Minimized]</code>	Auto-pauses playback when the player window is minimized
<code>[Video]</code>	Enables live video playback progress on the Windows taskbar
<code>[Image]</code>	Sets infinite display duration and enables cursor-centered zoom
<code>[night-audio]</code>	Dynamic audio normalizer balancing dialogues and sound effects
<code>[dolby-vision-perf]</code>	Low-overhead tone-mapping profile for smooth 4K DV/HDR playback